STACKS, DISCRIMINANTS: LEMMA-NORM-DIFFERENT-IS-DISCRIMINANT

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 1941-1955 of sources/stacks-discriminant/pinned-original-source.tex; proof span is lines 1957-1989, the complete author proof (31 body lines).

Editorial note. Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 14 of queue_discriminant.json (pb=1502, stmt 1941-1955, proof 1957-1989).

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original lemma 01X0: sources/stacks-discriminant/pinned-original-source.tex, source lines 1989–1989. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BW9: sources/stacks-discriminant/pinned-original-source.tex, source lines 1907–1912. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original lemma 0BW8: sources/stacks-discriminant/pinned-original-source.tex, source lines 1863–1867. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *Let $f : Y \to X$ be a flat quasi-finite morphism of Noetherian schemes. Let $R \subset Y$ be the closed subscheme defined by $\mathfrak{D}_f$.*

(1) *If $\omega_{Y/X}$ is invertible, then R is a locally principal closed subscheme of Y.*
(2) *If $\omega_{Y/X}$ is invertible and f is finite, then the norm of R is the discriminant D_f of f.*
(3) *If $\omega_{Y/X}$ is invertible and f is étale at the associated points of Y, then R is an effective Cartier divisor and there is an isomorphism $\mathcal{O}_Y(R) = \omega_{Y/X}$.*

Date: stacks-project:discriminant.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. Proof of (1). We may work locally on Y, hence we may assume $\omega_{Y/X}$ is free of rank 1. Say $\omega_{Y/X} = \mathcal{O}_Y\lambda$. Then we can write $\tau_{Y/X} = h\lambda$ and then we see that R is defined by h, i.e., R is locally principal.

Proof of (2). We may assume $Y \to X$ is given by a finite free ring map $A \to B$ and that $\omega_{B/A}$ is free of rank 1 as B-module. Choose a B-basis element λ for $\omega_{B/A}$ and write $\text{Trace}_{B/A} = b \cdot \lambda$ for some $b \in B$. Then $\mathfrak{D}_{B/A} = (b)$ and D_f is cut out by $\det(\text{Trace}_{B/A}(b_ib_j))$ where $b_1, \ldots, b_n$ is a basis of B as an A-module. Let $b_1^\vee, \ldots, b_n^\vee$ be the dual basis. Writing $b_i^\vee = c_i \cdot \lambda$ we see that $c_1, \ldots, c_n$ is a basis of B as well. Hence with $c_i = \sum a_{il}b_l$ we see that $\det(a_{il})$ is a unit in A. Clearly, $b \cdot b_i^\vee = c_i \cdot \text{Trace}_{B/A}$ hence we conclude from the computation in the proof of Lemma 0BW8 that $\text{Norm}_{B/A}(b)$ is a unit times $\det(\text{Trace}_{B/A}(b_ib_j))$.

Proof of (3). In the notation above we see from Lemma 0BW9 and the assumption that h does not vanish in the associated points of Y, which implies that h is a nonzerodivisor. The canonical isomorphism sends 1 to $\tau_{Y/X}$, see Divisors, Lemma 01X0. □